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HANDGUN HOLSTER WITH RETENTION MECHANISM

Abstract

A holster includes a retention catch arranged to enter a portion of the weapon to be in a locked position. A release lever is coupled to the holster housing, which includes a shaft arranged to push against a catch interface member that can move the retention catch out of the locked position. The release lever includes a tab which in an initial position rests on a first surface in the holster housing. The release lever is movable so that the tab moves towards a channel formed in the holster housing and the release lever is movable to overcome a biasing force of a biasing device so that the release lever enters into the channel and pushes against the catch interface member to move the retention catch out of the locked position into an unlocked position. Without application of any force on the release lever, the biasing device urges the tab against a second surface and the release lever cannot return to the initial position.

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Background/Summary

FIELD OF THE INVENTION
[0001] The present invention relates to handgun holsters with a retention mechanism.
BACKGROUND OF THE INVENTION
[0002] Many users of handguns, particularly military and law enforcement personnel, carry a handgun in a holster designed to protect the handgun and hold it securely. Holsters can be worn in a number of ways, such as on a belt at the waist, on the thigh, under an arm, or around an ankle.
[0003] Some holsters include a variety of strap or flap arrangements that prevent the removal of the firearm from the holster while the strap or flap is in place. The user must first unfasten and/or rotate the strap/flap before the firearm can be withdrawn. To re-secure the handgun in the holster once the handgun has been re-holstered, the user must physically refasten and/or rotate the strap/flap before the firearm is securely retained within the holster. Some users might not prefer these designs because of the time required to release and/or re-secure the handgun.
[0004] Accordingly, a user-friendly holster retention mechanism is needed.
SUMMARY
[0005] The present invention seeks to provide user-friendly holster retention mechanism, as is described more in detail hereinbelow.
[0006] There is thus provided in accordance with a non-limiting embodiment of the invention a holster including a holster housing which has an opening for inserting therethrough a weapon, and including a retention catch arranged to enter a portion of the weapon to be in a locked position, and a release lever coupled to the holster housing, the release lever including a shaft arranged to push against a catch interface member, the catch interface member being configured to move the retention catch out of the locked position, wherein the release lever includes a tab which in an initial position rests on a first surface (e.g., a first side of a shelf surface) in the holster housing, the release lever being movable so that the tab moves towards a channel formed in the holster housing and the release lever is movable to overcome a biasing force of a biasing device so that the release lever enters into the channel and pushes against the catch interface member to move the retention catch out of the locked position into an unlocked position, and in the unlocked position, without application of any force on the release lever, the biasing device urges the tab against a second surface (e.g., a second side of the shelf surface or some other surface in the holster housing) and the release lever cannot return to the initial position.
[0007] In accordance with a non-limiting embodiment of the invention a hood is arranged to pivot over a portion of the weapon, wherein in the initial position, the release lever prevents movement of the hood and wherein movement of the tab towards the channel releases and allows movement of the hood.
[0008] In accordance with a non-limiting embodiment of the invention an optics cover is arranged to pivot over a portion of the weapon, wherein in the initial position, the release lever prevents movement of the optics cover and wherein movement of the tab towards the channel releases and allows movement of the optics cover.
[0009] In accordance with a non-limiting embodiment of the invention a hood is arranged to pivot over a first portion of the weapon and an optics cover is arranged to pivot over a second portion of the weapon, wherein in the initial position, the release lever prevents movement of the hood and the optics cover and wherein movement of the tab towards the channel releases and allows movement of the hood and the optics cover. The hood may be coupled to the optics cover so that the hood and the optics cover move together. The hood may be coupled to the optics cover by means of link arms

formed with pivot axles that move in arcuate tracks formed in the optics cover. Alternatively, the hood may be coupled to the optics cover by magnetic attraction.

[0010] In accordance with a non-limiting embodiment of the invention the hood is formed with a recess and the tab is received in the recess in the initial position.

[0011] In accordance with a non-limiting embodiment of the invention the portion of the weapon in which the retention catch is arranged to enter is an ejection port or a trigger guard.

[0012] There is provided in accordance with a non-limiting embodiment of the invention a holster including a holster housing which has an opening for inserting therethrough a weapon, and including a retention catch arranged to enter a portion of the weapon to be in a locked position, a release lever coupled to the holster housing, the release lever including a shaft arranged to push against a catch interface member, the catch interface member being configured to move the retention catch out of the locked position, and a retention-level member that has two mounting positions in the holster housing, the retention-level member cooperating with the release lever so that at one of the mounting positions the release lever has level-one retention and at the other of the mounting positions the release lever has level-two retention.

[0013] In accordance with a non-limiting embodiment of the invention the release lever includes a movable pusher member arranged to move against a biasing force of a biasing device.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] The present invention will be understood and appreciated more fully from the following detailed description, taken in conjunction with the drawings in which:

[0015] FIG. **1** is a simplified perspective illustration of a holster, in accordance with a non-limiting embodiment of the present invention, in which a release lever is in a retaining position that retains a hood and optics cover in the locked position, and is also in a retaining position that retains a retention catch in a locked position in either an ejection port or trigger guard of a weapon.

[0016] FIG. **2** is a simplified perspective illustration of the holster, in which the release lever has been moved out of the hood and optics cover retaining position so that the hood and optics cover move together out of the locked position into an unlocked position, but the release lever is still in the retaining position that retains the retention catch in the locked position in the ejection port or trigger guard of the weapon.

[0017] FIG. **3** is a simplified perspective illustration of the release lever assembly.

[0018] FIGS. **4**A, **4**B and **4**C are perspective illustrations of three different positions of the release lever assembly, in which FIG. **4**A is the totally locked position in which the hood and optics cover are in the locked position and the retention catch for the ejection port or trigger guard is in the locked position, FIG. **4**B is the semi-locked position in which the hood and optics cover are unlocked but the retention catch remains locked (this first movement from totally locked to semi-locked, which is a horizontal and/or a pivoting movement, is level-one retention), and FIG. **4**C is the fully unlocked position in which the hood and optics cover and the retention catch are all unlocked (this second movement from semi-locked to fully unlocked, which is a vertical movement, is level-two retention). FIG. **4**C shows the resting position of the release lever after having been moved to the fully unlocked position, showing that the release lever cannot move back to the semi-locked position (and of course not back to the fully locked position) unless moved by the user horizontally to clear the obstacle that prevents upward movement of the release lever.

[0019] FIGS. **5**A and **5**B are respectively simplified perspective illustrations of the optics cover and link arms that couple the hood with arcuate tracks in the optics cover so that the optics cover moves with the hood.

[0020] FIGS. **5**C and **5**D are respectively simplified perspective illustrations of the optics cover with magnets and magnetic elements in the hood that are magnetically attracted to each other so that the optics cover moves with the hood.

[0021] FIG. **5**E is a simplified perspective illustration of the optics cover magnetically attracted to the hood.

[0022] FIGS. **6**A and **6**B are simplified front and side perspective illustrations, respectively, of the release lever engaged with the retention catch which is locked in the ejection port of the weapon.

[0023] FIG. **6**C is a simplified perspective illustration of the release lever about to push against a portion of the retention catch mechanism in order to pivot the retention catch out of the locked engagement in the ejection port of the weapon.

[0024] FIGS. **7**A and **7**B are simplified side and front perspective illustrations, respectively, of the release lever engaged with the retention catch which is locked in the trigger guard of the weapon.

[0025] FIG. **7**C is a simplified perspective illustration of the release lever about to push against a portion of the retention catch mechanism in order to pivot the retention catch out of the locked engagement in the trigger guard of the weapon.

[0026] FIG. **8** is a simplified perspective illustration of a holster, in accordance with a non-limiting embodiment of the present invention, in which a retention-level member cooperates with a release lever so that the user can select if the release lever has level-one or level-two retention.

[0027] FIG. **9** is a simplified perspective illustration of the retention-level member cooperates and the release lever, showing that the retention-level member has two assembly positions, one for level-one retention and the other for level-two retention.

[0028] FIG. **10** is an exploded illustration of the release lever showing a pushing element biased by a biasing device, such as a coil spring.

[0029] FIGS. **11**A and **11**B are perspective illustrations, respectively, of the retention-level member mounted at the level-one retention position, in which the release lever has to be moved in one movement (vertically) to release any catch holding the weapon, and the retention-level member mounted at the level-two retention position, in which the release lever has to be moved in two movements (horizontally and then vertically) to release any catch holding the weapon.

[0030] FIGS. **12**A and **12**B are perspective illustrations, respectively, of the holster with the retention-level member mounted at the level-one retention position, in which the release lever has to be moved in one movement (vertically) to release any catch holding the weapon, and the retention-level member mounted at the level-two retention position, in which the release lever has to be moved in two movements (horizontally and then vertically) to release any catch holding the weapon.

DETAILED DESCRIPTION

[0031] Reference is now made to FIGS. **1** and **2**, which illustrate a holster **10**, in accordance with a non-limiting embodiment of the present invention.

[0032] Holster **10** may include a holster housing **12**, made of any suitable holster material, such as but not limited to, a polymer, such as KYDEX (thermoplastic acrylic-polyvinyl chloride). The holster housing **12** has an opening **14** for inserting therethrough a weapon **15**. The holster housing **12** may include a paddle hub **16** (which may be serrated), or any other garment attachment means, as is known in the art.

[0033] Holster may include a hood **18** and an optics cover **20**. As seen in FIG. **1**, when in the locked position, hood **18** lies over a rear portion of the weapon **15**, such as but not limited to, the rear face of the slide, and optics cover **20** lies over a portion of the weapon where an optics accessory may be mounted, such as but not limited to, a laser aiming accessory mounted on a Picatinny rail on the upper side of the slide. As will be described below, hood **18** and optics cover **20** move together and are actuated by a release lever **22**.

[0034] Reference is now made to FIG. **3**, which illustrates the assembly of the release lever **22**. Release lever **22** may include a shaft **24** which has a pusher member **25** at an upper end thereof. The terms “upper” and “lower” are non-limiting relative terms, and refer to the positions shown in the drawings. A lower end **26** of shaft **24** is arranged to push against a catch interface member **27**. As will be described below, the catch interface member **27** is configured to move a retention catch **60** (seen partially in FIGS. **1** and **2**) out of a locked position in either an ejection port or trigger guard (catch **70**, FIG. **7**A) of the weapon.

[0035] A tab **28** protrudes sideways from shaft **24** at a position intermediate pusher member **25** and lower end **26** of shaft **24**. A biasing device **29**, such as a coil spring, provides a biasing force against shaft **24**. Biasing device **29** is placed against a counterforce surface **30**. The biasing force urges release lever **22** to be in the locked position, until the release lever **22** is moved by the user to an unlocked position.

[0036] Reference is now made to FIGS. **4**A, **4**B and **4**C are three different positions of the release lever **22**. In FIG. **4**A, the release lever **22** is in the totally locked position in which tab **28** lies over a shelf **32** of holster housing **12**. This is the position shown in FIG. **1**. As seen in FIG. **1**, in this totally locked position, hood **18** and optics cover **20** are in the locked position. As will be explained below, the retention catch for the ejection port or trigger guard is also in the locked position. As seen in FIGS. **1** and **2**, hood **18** may be formed with a recess **34**. In the locked position of FIG. **1**, tab **28** is received in recess **34** and thus prevents hood **18** (and optics cover **20**) from moving out of the locked position.

[0037] In FIG. **4**B, release lever **22** has been moved to the right (in the sense of the drawings) which moves tab **28** out of recess **34** of hood **18** (as seen in FIG. **2**). This is the semi-locked position in which the hood **18** and optics cover **20** are unlocked. This first movement is a movement from the totally locked position to the semi-locked position, which is a horizontal and/or a pivoting movement, and provides level-one retention. In this position, the retention catch remains locked, as explained below.

[0038] FIG. **4**C is the fully unlocked position in which the hood and optics cover and the retention catch are all unlocked. This is accomplished by moving the release lever **22** vertically downwards (in the sense of the drawings) into a channel **36** formed in holster housing **12**. This second movement from the semi-locked position to the fully unlocked position provides level-two retention. FIG. **4**C shows the resting position of the release lever **22** after having been moved to the fully unlocked position, showing that the release lever **22** cannot move back to the semi-locked position (and of course not back to the fully locked position) unless moved by the user horizontally to clear the underside of shelf **32**, which is the obstacle that prevents upward movement of the release lever **22**.

[0039] Reference is now made to FIGS. **5**A and **5**B, which illustrate the optics cover **20** and link arms **38** that couple the hood **18** (not shown here, but shown in FIGS. **1** and **2**). Optics cover **20** may be formed with arcuate tracks **40** and a pivot **42**. Each of the link arms **38** may be formed with a pivot axle **44**, and with a mounting provision **46**, such as a mounting hole.

[0040] As seen in FIGS. **1** and **2**, the pivot axle **44** slides in arcuate track **40**. Hood **18** pivots about a hood pivot **47**. Moving release lever from the locked position of FIG. **1** to the unlocked position of FIG. **2** allows hood **18** and optics cover **20** to move together and pivot downwards to the unlocked position. A biasing device (not shown), such as a coil spring at the pivot **32** or **47** or other suitable position, may urge the hood **18** and optics cover **20** to move to the unlocked position when release lever **22** is moved out of the way. The user may manually move hood **18** and optics cover **20** back to their original positions, by simply pushing them back to the original positions against the force of the biasing device that urged them to the open positions.

[0041] It is noted that the holster can be provided with hood **18** and without the optics cover **20**, or with the optics cover **20** and without the hood **18**.

[0042] Reference is now made to FIGS. **5**C and **5**D, which show an alternative way of hood **18** moving together with optics cover **20**. In this variation, as seen in FIG. **5**C, optics cover **20** may be provided with a magnet **48**, which may be made of any suitable magnetic material, such as but not limited to, rare-earth materials, e.g., neodymium iron boron or samarium cobalt and the like, or non-rare-earth materials, e.g., different ferrous alloys. The magnet **48** may be mounted in an insert **50**.

[0043] As seen in FIG. **5**D, hood **18** may be provided with a magnetic element **52** mounted in a crevice **54** formed in hood **18**. The magnetic element **52** may be made of a magnetic ferrous alloy, such as various steel alloys. As seen in FIG. **5**E, the magnet **48** of optics cover **20** and the magnetic element **52** of hood **18** are magnetically attracted to each other so that the optics cover **20** moves with the hood **18**.

[0044] Reference is now made to FIGS. **6**A and **6**B, which illustrate the release lever **22** engaged with the retention catch **60** which is locked in an ejection port **13** of weapon **15**. The retention catch **60** may be shaped as an arcuate lever that pivots about a lever pivot **62**. One end of retention catch **60** is an inclined interface surface **64** that interfaces with catch interface member **27** of release lever **22**. The opposite end of retention catch **60** is a tongue **66** that enters the ejection port **13**.

[0045] Reference is now made to FIG. **6**C. Release lever **22** may be moved downwards (arrow **61**) which causes catch interface member **27** to move sideways (arrow **63**) on inclined interface surface **64**. The downward movement is possible only when release lever has been moved to the position of FIG. **4**B and **4**C when it can then be pushed downwards in channel **36** (FIG. **4**C). The downward movement causes inclined interface surface **64** to move inwards (arrow **65**) and pivot the tongue **66** of retention catch **60** out of the locked engagement in the ejection port **13** of the weapon **15**. A biasing device **67**, such as a coil spring, urges catch interface member **27** upwards against release lever **22** so as to normally maintain tongue **66** of retention catch **60** in the locked engagement in ejection port **13**.

[0046] Reference is now made to FIGS. **7**A and **7**B, which illustrate the release lever **22** engaged with a retention catch **70** which is locked in an trigger guard **11** of weapon **15**. The retention catch **70** may be shaped as an arcuate lever that pivots about a lever pivot **72**. One end of retention catch **70** is an inclined interface surface **74** that interfaces with a catch interface member **27**A of release lever **22**. The opposite end of retention catch **70** is a tongue **76** that enters the trigger guard **11**.

[0047] Reference is now made to FIG. **7**C. Release lever **22** may be moved downwards (arrow **71**) which causes catch interface member **27**A to move downwards (arrow **73**) on inclined interface surface **74**. Again, the downward movement is possible only when release lever has been moved to the position of FIG. **4**B and **4**C when it can then be pushed downwards in channel **36** (FIG. **4**C). The downward movement causes inclined interface surface **74** to move inwards (arrow **75**) and pivot the tongue **76** of retention catch **70** out of the locked engagement in the trigger guard **11** of the weapon **15**. A biasing device **77**, such as a coil spring, urges catch interface member **27**A upwards against release lever **22** so as to normally maintain tongue **76** of retention catch **70** in the locked engagement in trigger guard **11**.

[0048] Reference is now made to FIG. **8**, which illustrates a holster **80**, in accordance with a non-limiting embodiment of the present invention. Holster **80** may include a holster housing made similar to holster **10**. Holster **80** includes a retention-level member **82** that cooperates with a release lever **84** so that the user can select if the release lever **84** has level-one or level-two retention, as is now described.

[0049] Reference is now made additionally to FIG. **9**. The retention-level member **82** may include an elongate member **86** that has a grasping member **88** at one portion thereof and a position selection member **90** at another portion thereof. The position selection member **90** may be formed with two mounting holes **92** and **94** (FIG. **9**). As seen in FIG. **8**, position selection member **90** may be received in a channel housing **96** in holster **80**. A fastener **98**, such as a screw, may secure position selection member **90** to channel housing **96** at either mounting hole **92** or **94**.

[0050] As seen in FIG. **9**, release lever **84** may be arranged to operate retention catch **60** or **70**, as described above. Release lever **84** differs from release lever **22** in that release lever **84** has a movable pusher member **85** which slides along a head **81** and is biased by a biasing device **83** (FIG. **10**).

[0051] Reference is now made to FIGS. **11**A and **12**A, which illustrate the retention-level member **82** mounted at the level-one retention position, in which the release lever **84** has to be moved in one movement (vertically downwards) to release any catch holding the weapon.
[0052] Reference is now made to FIGS. **11**B and **12**B, which illustrate the retention-level member **82** mounted at the level-two retention position, in which the release lever **84** has to be moved in two movements (horizontally and then vertically) to release any catch holding the weapon. In this position, movable pusher member **85** is blocked from downwards movement because it abuts against the top of position selection member **90**. The movable pusher member **85** must first be moved horizontally against the force of the biasing device **83** (FIG. **10**) (this is the first movement) and only then the release lever **84** can be moved downwards (the second movement) to release any catch holding the weapon.
[0053] It is noted that retention-level member **82** and release lever **84** may be incorporated in the embodiment of holster **10**. If such an embodiment is used with the hood **18**, when the retention-level member **82** is mounted for level-two retention of the release lever **84**, the addition of hood **18** makes the holster have a total of level-three retention. This is because the hood **18** also protects the handgun from being removed from the holster and the hood **18** cannot be released unless the user first moves movable pusher member **85** horizontally against the force of the biasing device **83** (FIG. **10**). In such an embodiment, after the user moves movable pusher member **85** horizontally, then release lever **84** can be moved down in the channel **36** (FIG. **4**C) to release hood **18** (and optics cover **20**, if provided, which moves together with hood **18**) and to release the catches in the ejection port or trigger guard.

Claims

1. A holster comprising: a holster housing which has an opening for inserting therethrough a weapon, and comprising a retention catch arranged to enter a portion of the weapon to be in a locked position; and a release lever coupled to said holster housing, said release lever comprising a shaft arranged to push against a catch interface member, said catch interface member being configured to move said retention catch out of said locked position; wherein said release lever comprises a tab which in an initial position rests on a first surface in said holster housing, said release lever being movable so that said tab moves towards a channel formed in said holster housing and said release lever is movable to overcome a biasing force of a biasing device so that said release lever enters into said channel and pushes against said catch interface member to move said retention catch out of said locked position into an unlocked position, and in said unlocked position, without application of any force on said release lever, said biasing device urges said tab against a second surface and said release lever cannot return to said initial position.
2. The holster according to claim 1, further comprising a hood arranged to pivot over a portion of the weapon, wherein in said initial position, said release lever prevents movement of said hood and wherein movement of said tab towards said channel releases and allows movement of said hood.
3. The holster according to claim 1, further comprising an optics cover arranged to pivot over a portion of the weapon, wherein in said initial position, said release lever prevents movement of said optics cover and wherein movement of said tab towards said channel releases and allows movement of said optics cover.
4. The holster according to claim 1, further comprising a hood arranged to pivot over a first portion of the weapon and an optics cover arranged to pivot over a second portion of the weapon, wherein in said initial position, said release lever prevents movement of said hood and said optics cover and wherein movement of said tab towards said channel releases and allows movement of said hood and said optics cover.
5. The holster according to claim 4, wherein said hood is coupled to said optics cover so that said hood and said optics cover move together.

6. The holster according to claim 5, wherein said hood is coupled to said optics cover by means of link arms formed with pivot axles that move in arcuate tracks formed in said optics cover.
7. The holster according to claim 5, wherein said hood is coupled to said optics cover by magnetic attraction.
8. The holster according to claim 2, wherein said hood is formed with a recess and said tab is received in said recess in said initial position.
9. The holster according to claim 1, wherein the portion of the weapon in which said retention catch is arranged to enter is an ejection port or a trigger guard.
10. The holster according to claim 1, further comprising a retention-level member that has two mounting positions in said holster housing, said retention-level member cooperating with said release lever so that at one of said mounting positions said release lever has level-one retention and at the other of said mounting positions said release lever has level-two retention.
11. The holster according to claim 10, wherein said release lever comprises a movable pusher member arranged to move against a biasing force of a biasing device.
12. A holster comprising: a holster housing which has an opening for inserting therethrough a weapon, and comprising a retention catch arranged to enter a portion of the weapon to be in a locked position; a release lever coupled to said holster housing, said release lever comprising a shaft arranged to push against a catch interface member, said catch interface member being configured to move said retention catch out of said locked position; and a retention-level member that has two mounting positions in said holster housing, said retention-level member cooperating with said release lever so that at one of said mounting positions said release lever has level-one retention and at the other of said mounting positions said release lever has level-two retention.
13. The holster according to claim 12, wherein said release lever comprises a movable pusher member arranged to move against a biasing force of a biasing device.